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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part - III Examination, 2024
Course Title: Inorganic Chemistry (IV) (Selected Topics)
Course No: Chem-322
Full Marks: 35, Time: 2.5 hours

(Answer Any Five Questions)

1. (a) Distinguish among conductors, semiconductors and insulators on the basis of band theory of solids. 3
   (b) As the temperature increases the electrical resistance of a metal increases – explain. 2
   (c) What are intrinsic and extrinsic semi conductors? 2

2. (a) Categorize the followings into protic, aprotic and coordinating solvents.
   (i) HF, (ii) BrF_3, (iii) MeOH, (iv) Et_2O, (v) Toluene, (vi) MeCN, (vii) Me_2CO 2
   (b) Is it possible to differentiate between the acid strengths of HCl, HBr, and HI in liquid ammonia? Why? 2
   (c) Write down equations for the autoionization of NH_3, N_2O_4, SO_2, BrF_3, HF, and HSO_3F. 3

3. (a) Draw the structures of B_5H_9 and $B_{10}H_{14}$, and discuss the nature of unusual bonding present in diborane (B_2H_6). 1+2=3
   (b) "Boron hydrides are high potential hydrogen storage materials"– Justify the statement. 1
   (c) What happens when diborane reacts with? 2
   (i) Excess ammonia at low temperature
   (ii) Alcohol
   (d) Explain why boric acid (H_3BO_3) is a weak monobasic acid, but behaves as a strong acid in the presence of a 1,2-diol compound. 1

4. (a) Describe the synthesis of silicones from alkyl or aryl chlorosilanes. Explain the role of hydrolysis and condensation steps. 2
   (b) Describe the preparation and structure of phosphazenes. Explain their bonding and properties. 3
   (c) Write short notes on: 2
   (i) Silicones as lubricants and sealants
   (ii) Phosphazenes as high-performance polymers

5. (a) Hemoglobin acts as oxygen carrier whereas cytochrome acts as electron carrier although the active site for both of them are same. Explain why? 2
   (b) Draw and compare the oxygen saturation (binding) curves of myoglobin and hemoglobin with respect to oxygen partial pressure. 3
   (c) (i) What is chlorophyll? 1
   (ii) What is the function of chlorophyll in biological systems? 1

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6. (a) Discuss the variation in the ionic radii between La^{3+} and Lu^{3+}. 2
   (b) Explain why UF_3 and UF_4 are high melting point solids whereas UF_6 sublimes at only 57° C. 2
   (c) The electronic spectra of lanthanoid complexes contain many absorptions some of which are weak and sharp and similar to those of the gas-phase metal ions, and some of which are broad and are affected by the ligands present. What is your comments for this observation? 3

7. (a) "$[B_6H_6]$ is readily reduced by two electrons" - Explain with the help of Wade's rule. 2
   (b) What is the leveling effect? Explain it with reference to water and liquid ammonia. 2
   (c) Draw the structures of Fe_2S_2, Fe_4S_4, and Fe_3S_4 type iron sulfur co-factors. 3

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part - III Examination, 2023
Course Title: ***Inorganic Chemistry* (IV) (*Selected Topics*)**
Course No: Chem-322
Full Marks: 35, Time: 2.5 hours

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(***Answer Any <u>Five</u> Questions***)

1. (a) Describe the structure and bonding in pure metal and alloys. Explain the properties of pure metals and alloys using your understanding of their structure and bonding. 2+2=4
(b) Describe the band theory of metal with examples. 3

2. (a) Compare the solvent properties of H_2O and liquid NH_3. 2
(b) Discuss the properties of ionizing solvent. 2
(c) Write the self ionization reactions of liquid SO_2 and liquid NH_3 with examples 2
(d) Which of the following non-aqueous solvents has the longest liquid range?
(i) H_2SO_4 (ii) N_2O_4 (iii) NH_3 (iv) HF and why? 1

3. (a) Define electron deficient compound with examples. Compare the structures of C_2H_6 and B_2H_6 compounds. 1+2=3
(b) Write the reactions of B_2H_6 with NH_3 at different conditions 2
(c) Draw the structures of the following compounds:
(i) $Al(BH_4)_3$ and (ii) $Be(BH_4)_2$. 2

4. (a) Explain why the 4f electrons of the lantnanide elements do not take part in chemical bonding. 2
(b) Discuss the methods for the separation of lanthanides from Monazite. 3
(c) Discuss the color of actinide ions. 1
(d) How do the lanthanide complexes act as NMR shift reagent? 1

5. (a) Explain the similarities and differences in structure between myoglobin and hemoglobin in the deoxy and oxy state. 2
(b) Why are the distal and proximal histidines in myoglobin or hemoglobin. 1
(c) What is the oxyhemoglobin dissociation curve? Draw the Standard Oxyhemoglobin Dissociation Curve and explain the factors that affect the standard dissociation curve. 3

6. (a) What is called polymer? How is polymer classified? 1+2+1= 4
(b) How are $[NPCl_2]_3$ and $[NPCl_2]_4$ produced? 1.5
(c) Discuss the salient feature of structure and bonding in $[NPCl_2]_3$. 1.5
(d) Draw the formulae of one polymer having Si–O, B–N and S–S bond. 2

7. Write short notes on the following topics: (any three) 3+2+2=7
(i) Acid-base phenomenon in non-aqueous solvent
(ii) Sulfur-nitrogen polymer
(iii) Chemistry of Fullerenes
(iv) Iron-sulfur proteins
(v) Vitamin B_{12} co-enzymes

Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2022
Course Title: Inorganic Chemistry IV (Selected Topics)
Course No.: Chem-322H
Full Marks: 35, Time: 2.5 hours

Answer any five (5) questions

1. (a) Write down the physical and chemical properties that are associated with metals. 2
(b) Explain why the electrical conductivity of a metal decreases as the temperature is raised but the opposite occurs with semiconductors. 2
(c) What is meant by interstitial and substitutional alloys? Draw the structures for both metal and alloys. 2 + 1 = 3

2. (a) Define nonaqueous solvents with examples. How liquid ammonia enhances the acidity of compounds that are weak in water? 1 + 2 = 3
(b) Show the following reactions both in liquid ammonia and water:
(i) Self-ionization, (ii) Amphoteric behavior of $[Zn(OH)_2]$ 2
(c) Anhydrous H_2SO_4 is highly viscous and has low ion mobilities – explain? 2

3. (a) Discuss the unusual bonding in diborane. Draw the structures of B_5H_9 and $B_{10}H_{14}$. 2 + 2 = 4
(b) Draw the structure of borazine and explain its non-aromatic character. 2 + 2 = 4

4. (a) When $[Al(BH_4)_3]$ is heated at 70 °C, $[Al_2B_4H_{18}]$ is obtained. Write the equation for this reaction and draw the structure of this product. 3
(b) Although boric acid is a weak monobasic acid, it behaves as a strong acid in the presence of a 1,2-diol -explain. 1

5. (a) What is silicone polymer? 2
(b) How are phosphazene polymers obtained? Describe the bonding of phosphazene. 2
(c) What happens when $[NPCl_2]_3$ is heated to 250 °C followed by treatment with (i) RONa and (ii) RNH_2. 2
(d) Give the synthesis of S_4N_4? How does $(SN)_x$ behave like a one-dimensional conductor. 2

6. (a) Hemoglobin acts as oxygen carrier whereas cytochrome acts as electron carrier although the active sites of both of them are same – explain why? 2
(b) Show the oxygen saturation curves for myoglobin and hemoglobin against oxygen pressure. 2
(c) What is Chlorophyll? What is the function of Chlorophyll? 1
(d) Draw the structure of the heme group present in hemoglobin and myoglobin. 2

7. (a) Of all $Ln(OH)_3$ compounds, basicity is maximum for $Ce(OH)_3$ and that of $Lu(OH)_3$ is minimum – explain why? Complete the following reaction and write the formulae of the products: $Lu(OH)_3 + NaOH \rightarrow$ 2 + 1 = 3
(b) All but Lu^{3+} of the Lanthanide ions show absorption in the visible or near UV regions of the spectrum and their colors are pale in appearance – explain. 1
(c) Ce^{3+} and Yb^{3+} are colorless but show strong absorption in UV region – explain. 1
(d) What are chemical shift reagents? Give two examples. How chemical shift reagents work? 2

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part - III Examination, 2021
Course Title: Inorganic Chemistry (IV) (Selected Topics)
Course No: Chem-322
Full Marks: 35, Time: 2.5 hours

(Answer Any Five Questions)

1. (a) Explain the following properties of metal in term of electron-pool or electron gas theory:
(i) Electrical conductivity.
(ii) Malleability
(iii) Ductility. 3
(b) As the temperature increases the electrical resistance of a metal increases – explain. 2
(c) What are called Fermi Level and Forbidden Zones? 2

2. (a) What are non-aqueous, protic and aprotic solvents? 1.5
(b) Mention some properties of ionizing solvents. 1.5
(c) The solution of alkali metals in liquid ammonia is paramagnetic and blue in colour. – Explain. 1.5
(d) Acetic acid is a weak acid in aqueous solution but behaves as strong acid in liquid NH_3 – explain why. 1
(e) Write a brief note on ionic liquid as solvent. 1.5

3. (a) Explain why boric acid is a weak monobasic acid but behaves as a strong acid in presence of a 1,2-diol compound? 2.5
(b) What happens when diborane reacts with- 3
(i) Excess ammonia at low temperature (ii) H_2O (iii) Alcohol
(c) "Boron hydrides are high potential hydrogen storage materials"– Justify the statement. 1.5

4. (a) How does BN resemble with and differ from graphite? Draw the formulae of one polymer having Si-O, B-N and S-S bond. 3
(b) What happens when $[NPCl_2]_3$ is heated to 250 ^{0}C and followed by treatment with
(i) CH_3ONa (ii) RNH_2 2
(c) Explain how fullerene C_{60} behaves as a ligand and cage organometallic complexes. 2

5. (a) Discuss the cooperative effect of heme group in hemoglobin and compare with that in myoglobin. 2
(b) Draw the structure of porphine and explain how the structures of heme and chlorophyll are related to it. 2
(c) Explain the roles of magnesium ion play in the functioning of chlorophyll. 1.5
(d) Write a brief note on vitamin B_{12}. 1.5

6. (a) Explain the following: 2
(i) Of all $Ln(OH)_3$ compounds, basic character of $Ce(OH)_3$ is maximum and that of $Lu(OH)_3$ is minimum.
(ii) Covalent radii of Mo (Z=42) and W (Z=74) are approximately the same. [1.29 Å and 1.30 Å respectively]
(b) Mention the fluorides of uranium and give reactions for their formation from UO_2. 2
(c) Draw the structure of the following complexes and mention the coordination number of the central metals present in them. (i) $[UO_2(CH_3COO)_3]^-$ and (ii) $[Th(NO_3)_6]^{2-}$. 2
(d) Of all the lanthanides Yb (Z=70) shows the highest ionization energy for the formation of +3 ion – why? 1

7. (a) What are intermetallic compounds? Mention their properties. 2
(b) What are borohydrides? Draw the structure of the compounds (i) $Be(BH_4)_2$, (ii) $Al(BH_4)_3$. 2
(c) What happens when - 2
(i) BiI_3 reacts with KNH_2 in Liq. NH_3.
(ii) KI and $SbCl_5$ are treated in Liq.SO_2.
(d) Liquid SO_2 solutions of both water and hydrogen bromide are nonconductors of electricity but a mixture of the two becomes a good conductor - Explain why. 1

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2020
Course Title: Inorganic Chemistry IV (Selected Topics)
Course # Chem. 322, Full Marks: 35, Time: 2.5 hrs.

[Answer any five questions]

1. (a) Mention some characteristic properties of metals. 1.0
   (b) Show the difference among conductors, semiconductors and insulators on the basis of band theory of solids. 3.0
   (c) What is called Fermi Level? What are the P and N-type semi conductors? 3.0

2. (a) Discuss the properties of Ionizing solvent. 2.0
   (b) Compare the solvent properties H_2O and liquid NH_3. 2.0
   (c) The dissolution of alkali metal in liq. NH_3 produce a
   (i) Blue Coloration (ii) is a good conductor of electricity – explain. 1.5+1.5

3. (a) What is lanthanide contraction? What is the consequence of lanthanide contraction? 1+1
   (b) Discuss the ion-exchange method for the separation of lanthanides. 2.0
   (c) What are the points of differences and similarities between the lanthanides and actinides? 1.5
   (d) How do the lanthanide complexes act as NMR shift reagent? 1.5

4. (a) Hemoglobin acts as oxygen carrier whereas cytochrome acts as electron carrier although the active site for both of them are same. Explain why? 2.0
   (b) Show the oxygen saturation curves, for myoglobin and hemoglobin against oxygen pressure. 3.5
   (c) What is the Function of Chlorophyll? 1.5

5. (a) What is called polymer? How is polymer classified? 1.5
   (b) How are $[NPCl_2]_3$ and $[NPCl_2]_4$ produced? 2.0
   (c) Discuss the salient feature of structure and bonding in $[NPCl_2]_3$. 2.0
   (d) Draw the formulae of one polymer having Si-O, B-N and S-S bond. 1.5

6. (a) Draw the structures of B_5H_9 and $B_{10}H_{14}$. 2.0
   (b) Discuss the Structure / Bonding element of B_2H_6. 3.0
   (c) How is borazine prepared? Borazine is Inorganic benzene – explain. 2.0

7. Write short notes (any three) of the following: 2.5+2.5+2
   (a) Metal Carbide
   (b) Electron Sea Model in Metal bonding
   (c) Electron deficient compound
   (d) Nitrogen Fixation
   (e) Fullerene-C^{60}
   (f) Solid solution.

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CHE-47

Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part-III Examination 2018
Course Title: Inorganic Chemistry IV (Selected Topics)
Course: Chem. 322
Full Marks: 70, Time: 4 hrs.

Answer any FIVE questions

1. (a) Explain the following terms with example:
(i) Plane of symmetry,
(ii) Center of symmetry,
(iii) Improper axis of rotation and
(iv) Symmetry operation 4
(b) Draw sketches to indentify the following symmetry elements-
(i) A σ_v and a σ_h in BF_3 (ii) A C_3 axis and a σ_v plane in $CHCl_3$ 6
(c) Use a flow chart to assign the point group to each of the following.
(i) BF_3, (ii) $CHCl_3$ and (iii) B_2H_6

2. (a) Discuss the main feature of Band Theory of metals. 4
(b) Show the difference among conductors, semiconductors and insulators on the basis of band theory of solids. 3
(c) What is called Fermi Level? What are the P and N-type semi conductors? 2+2
(d) As the temperature increases the electrical resistance of metal increases-explain. 3

3. (a) What is Silicone polymer? Discuss the preparation of Silicones. 1+2
(b) How are $[NPCl_2]_3$ and $[NPCl_2]_4$ produced. Discuss the salient feature of structure and bonding in $[NPCl_2]_3$ 2+4
(c) What happens when $[NPCl_2]_3$ (i) is hydrolyzed with water, (ii) reacts with excess of NH_3, (iii) reacts with NaF and (iv) $NaOCH_3$. 2
(d) Silicone and phosphazene are isoelectronic-explain. 3

4. (a) What do you mean by essential and trace elements in biological process? 2
(b) What are meant by porphine and metalloporphyrins? 2
(c) Explain Deoxy-Mb is paramaganetic whearas Oxy-Mb is diamagnetic. 3
(d) Hemoglobin acts as oxygen carrier whereas cytochrome acts as electron carrier although the active site for both of them is same. Explain why? 3
(e) Show the oxygen saturation curves for myoglobin and hemoglobin against oxygen pressure. 4

5. (a) Discuss the methods for the separation of lanthanides from Monazite. 3
(b) Explain why the 4f electrons of the lanthanide elements do not take part in chemical bonding. 2
(c) Why La^{3+}, Ce^{4+} and Lu^{3+} are diamagnetic but all other lanthanides ions are paramagnetic-explain? 2
(d) What are the differences and similarities between the lanthanides and actinides? 2
(e) Actinides show higher oxidation state than lanthanides-explain? 2
(f) Why are Eu^{2+} and Yb^{2+} more stable than those of Eu^{3+} and Yb^{3+} ions? 2
(g) Why $Ce(OH)_3$ is more basic than $Lu(OH)_3$? 1

7. (a) What are the features of a substance to be a solvent?
(b) The solution of alkali metals in liquid ammonia is paramagnetic and the color of the solution is blue – explain. 4
(c) Explain the following with proper chemical equation: 3
(i) Solvolysis of Cl_2 in NH_3 is parallel to that in H_2O.
(ii) CH_3COOH a weak acid in H_2O but behaves as a strong acid in NH_3.
(iii) Urea shows no acidic behavior at all in H_2O but behaves as a strong acid in NH_3.
(d) Compare the properties of liquid NH_3 and H_2O as ionizing solvents. 2
(e) Complete the following reactions (any four): 4

(i) $PCl_5 + SO_2 \xrightarrow{\text{Liq. } SO_2}$

(ii) $FeCl_3 + KI \longrightarrow$

(iii) $HClO_4 + H_2SO_4 \xrightarrow{\text{Liq. } H_2SO_4}$

(iv) $CH_3COOH + H_2SO_4 \xrightarrow{\text{Liq. } H_2SO_4}$

(v) $NH_2CONH_2 + H_2SO_4 \xrightarrow{\text{Liq. } H_2SO_4}$

8. (a) Draw the structures of B_5H_9 and $B_{10}H_{14}$. Show by orbital diagram the formation of 3c-2e open and closed BBB bonds. 2+3
(b) Why borazine is called an inorganic benzene? 3
(c) What happens when diborane reacts with (i) excess ammonia at low temperature and (ii) water? 2
(d) "Boron hydrides are high potential hydrogen storage materials"-Justify the statement. 4

8. (a) Define the following terms: (i) Diamagnetism (ii) Ferromagnetism (iii) Antiferromagnetism 3
(b) What do you understand by temperature independent magnetism? 2
(c) What is Curie point? Explain the effect of temperature on susceptibility of ferromagnetic substance 5
(d) What is magnetic moment? Which of the following nuclei should have zero magnetic moment: $^{2}_{1}H$, $^{12}_{6}C$, $^{14}_{7}N$, and $^{16}_{8}O$ 4

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Department of Chemistry
Jahangirnagar University
B.Sc. (Hons.) Part- III Final Examination 2017
Course # Chem. 322
Full Marks: 35, Time: 2.5 hrs.

[Answer any Four questions]

1. (a) Mention three prime properties of metal. 0.75
(b) What are the major crystal structures observed in metals? 0.50
(c) (i) Define Heats of Atomization (ii) Explain the fact that Melting Point decreases Across the group Li-Na-K-Rb-Cs. 0.5+1
(d) Explain the MOT in Li_4 molecule. 2
(e) (i) Define alloys. 0.5
(ii) Give Hume-Rothery Rules those define the miscibility of metals in alloys. 1.25
(iii) Explain the fact that Li/Na or K/Na alloys don't exist. 1
(f) What is a Perovskite structure? Give the formula and draw the structure of a common Perovskite material. 1.25

2. (a) What are non-aqueous solvents? Name three non-aqueous solvents. 1
(b) Addition of Alkali metal to NH_3 creates an intense blue solution with electrical conductivity comparable to metal. Explain the following:
(i) 'Solvated electron' theory.
(ii) The blue solution is paramagnetic.
(iii) The blue solution is a good one-electron reducing agent. 0.75×3
(c) Explain the following with proper chemical equation:
(i) Solvolysis of Cl_2 in NH_3 is parallel to that in H_2O
(ii) CH_3COOH, a weak acid in H_2O but behaves as a strong acid in NH_3.
(iii) Urea shows no acidic behaviour at all in H_2O, but behaves as weak acid in NH_3. 0.5×4=2
(d) Complete the following reactions (any four):

(i) $PCl_5 + SO_2 \xrightarrow{liq.\ SO_2}$ $SOCl_2 + POCl_3$

(ii) $FeCl_3 + KI \xrightarrow{liq.\ SO_2}$ $FeI_2 + KCl + I_2$

(iii) $HClO_4 + H_2SO_4 \xrightarrow{liq.\ H_2SO_4}$ $H_3SO_4^+ + ClO_4^-$

(iv) $CH_3COOH + H_2SO_4 \xrightarrow{liq.\ H_2SO_4}$ $CH_3CO^+ + H_3SO_4^+$

(v) $NH_2-CONH_2 + H_2SO_4 \xrightarrow{liq.\ H_2SO_4}$ $^-NH-C(NH_2)_2 + H_3SO_4^+$

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(a) What is an electron deficient compounds? Show by orbital diagram the formation of B_2H_6. 4
(b) Explain the structure of S_4N_4 compound. 2.75

4. (a) What is an electron deficient compounds? Show by orbital diagram the formation of B_2H_6.
(b) Explain the structure of S_4N_4 compound.

5. (a) Explain the structural similarities and differences between haemoglobin in oxy-deoxy states. 2
(b) What are distal and proximal histidines? What are their role in oxygen transport? 2
(c) Draw structure of chlorophyll. 1.75
(d) What are cytochromes? Structurally how do they differ from haemoglobin? 2
(e) What is an apoprotein? 1

6. (a) What are lanthanides? Write down their common electronic configuration. 1.25
(b) What is Lanthanide contraction? State two consequences of lanthanide contraction. 2
(c) Name the various methods for extraction of lanthanides and discuss any one method for their extraction and separation. 3
(d) Explain the facts: 2×0.75
(i) Ln^{3+} ions do not form complexes readily like transition metals.
(ii) Ln^{3+} ions forms stable complexes with oxygen rich chelating ligands.

7. Write short note (any three) of the following: 3+3+2.75
(a) Metal carbides
(b) Fluorocarbons
(c) Nitrogen fixation
(d) Chemistry of fullerenes
(e) Vitamin B and B_{12} co-enzymes.

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Department of Chemistry
Jahangirnagar University
B. Sc. (Part III) Honours Examination 2016
Course Title: Inorganic Chemistry IV (Selected Topics)
Course No: Chem - 322
Time: 2.5 Hours, Total Marks: 35

Answer any four questions

1. (a) What are metals? Mention some characteristic properties of metals. — 2
   (b) Explain the band theory of metals. — 0.75+2
   (c) Based on band theory of metals clearly explain the following: — 4
   (i) Fermi level (ii) forbidden zone (iii) insulator and (iv) semiconductor — 1+1

2. (a) What are the properties of solvent? How are solvents classified? — 2
   (b) Addition of alkaline-earth metal in liq. ammonia produces a complex, whereas the solution of alkali metal in liq. ammonia is coloured-explain. — 2.75
   (c) Compare the properties of liquid NH_3 and H_2O as ionizing solvents. — 0.5×4=2
   (d) Complete the following reactions (any four):.

   (i) $KI + NH_4Cl \xrightarrow{liq.\ NH_3}$

   (ii) $NH_4Cl + KNH_2 \xrightarrow{liq.\ NH_3}$

   (iii) $BCl_3 + NH_3 \xrightarrow{liq.\ NH_3}$

   (iv) $PCl_5 + SO_2 \xrightarrow{liq.\ SO_2}$

   (v) $FeCl_3 + KI \xrightarrow{liq.\ SO_2}$

   (vi) $SbCl_3 + KCl \xrightarrow{liq.\ SO_2}$

3. (a) What is Lanthanide contraction? What are the consequences of Lanthanide contraction? — 2
   (b) Name the various methods for the extraction of Lanthanide and discuss one method for the extraction and separation of Lanthanide. — 3
   (c) State the colour of actinide ions. — 1.25
   (d) What are the similarities and differences between Lanthanide and Actinide elements? — 2.5

4. (a) Explain the similarities and differences in structures between myoglobin and hemoglobin in the deoxy- and oxy- states. — 2
   (b) Why are the distal and proximal histidines in myoglobin or hemoglobin? — 1
   (c) What is oxyhemoglobin dissociation curve? Draw the standard oxyhemoglobin Dissociation Curve and explain the factors that govern the dissociation curve. — 1+2+2
   (d) What are the structural features of Chlorophyll? — 0.75

5. (a) What is called polymer? How is polymer classified? — 1.75
   (b) How are $[NPCl_2]_3$ and $[NPCl_2]_4$ produced? — 2
   (c) Discuss the salient feature of structure and bonding in $[NPCl_2]_3$. — 3
   (d) Draw the formulae of one polymer having Si-O, B-N and S-S bond. — 2

6. (a) What are intermetallic compounds? Mention their properties — 2.5
   (b) Liquid SO_2 solutions of both water and hydrogen bromide are nonconductors of electricity but a mixture of the two becomes a good conductor-Explain why. — 1.75
   (c) What are borohydrides? Draw the structure of the compounds (i) $Be(BH_4)_2$, (ii) $Al(BH_4)_3$. — 2
   (d) Write a brief note on fluorocarbons. — 2.5

7. (a) Write short notes on **(any three)** of the following: — 3+3+2.75
   (a) Metal carbide
   (b) Vitamin B_{12}
   (c) Solar Cell
   (d) Hydrides of Borane and their usage as reducing agents
   (e) Fullerene-C_{60}

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Department of Chemistry
Jahangirnagar University
B. Sc. (Part III) Honours Examination 2015
Course Title: Inorganic Chemistry (IV) (Selected Topics)
Course No.: Chem - 322
Time: 2.5 Hours, Total Marks: 35

Answer any four questions

1. (a) Distinguish among conductors, semiconductors and insulators on the basis of band theory of solids. 3.0
(b) As the temperature increases the electrical resistance of a metal increases-explain. 2.0
(c) What is called doping? What are the p and n-type semiconductors? 2.75
(d) What is called supper lattice? 1.0

2. (a) Explain why the 4f electrons of the lanthanide elements do not take part in chemical bonding. 1.5
(b) Discuss the methods for the separation of lanthanides from monazite. 4.0
(c) Discuss the colour of actinide ions. 1.5
(d) Actinide show higher oxidation state than that of lanthanide-explain. 1.75

3. (a) Hemoglobin acts as oxygen carrier whereas cytochrome acts as electron carrier although the active site for both of them are same. Explain why? 2.0
(b) Show the oxygen saturation curves for myoglobin and hemoglobin against oxygen pressure. What are the factors that govern the saturation curves? 2+1 = 3
(c) Draw the structure of vitamin B_{12}. 2.0
(d) Sketch the block diagram of "Nitrogen Cycle." 1.75

4. (a) What is an electron deficient compound? Show by the orbital diagram the formation of 3C-2e open and closed BBB bonds. 2+2 =4
(b) Justify the borazine as an electronic isomer of benzene. 3.0
(c) Explain the order of acidity for the following series. 1.75
$BF_3 < BCl_3 < BBr_3$

5. (a) Draw the structures of B_5H_9 and $B_{10}H_{14}$ and discuss the unusual bonding in diborane. 1+1=2
(b) What happens when $[NPCl_2]_3$ is heated to 250°C and followed by treatment with (i) RONa and (ii) RNH_2 2.0

6. (a) Explain why boric acid is a weak monobasic acid but behaves as a strong acid in Presence of a 1,2-diol compound. 2.5
(b) "B-N bond in borazine is slightly polar but C-C bond in benzene is non-polar"-explain. 2.25
(c) What are the features of a substance to be a solvent? 1.0
(d) How are the solvents classified? 2.0
(e) The solution of alkali metals in liquid is paramagnetic and the colour of the solution is blue-explain. 2.0
(f) Acetic acid is a weak acid in H_2O solution but behaves as a strong acid in liquid NH_3 - explain. 2.0
(g) Separate solutions of liquid SO_2 in H_2O and liquid SO_2 in HBr cannot conduct electricity but mixture of the two solutions is a good conductor of electricity – explain. 1.75

7. (a) What are metalloporphyrins? Give one biological importance of metalloporphyrin. 2.0
(b) What is photosynthesis? What is the function of chlorophyll molecule in photosynthesis. 2.0
(c) What happens when biological systems containing hemoglobin are exposed to CO? 2.0
(d) Draw the structure of porphyrin and explain how the structures of heme and chlorophyll are related to it? 2.75

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